PROJECT TESTING PHASE

Project Title: Streamlining IT Procurement: Automating Standard Laptop Orders

Topic: Project Testing Phase

1. Introduction

The Testing Phase is an important stage of the project “Streamlining IT Procurement: Automating Standard Laptop Orders.” The main purpose of testing is to verify that the automated laptop ordering process works correctly, efficiently, and without errors. The complete process, from submitting a laptop request to approval and procurement processing, is tested before implementation.

2. Objectives of Testing

• To verify that users can successfully submit laptop requests.
• To check whether mandatory fields are validated correctly.
• To ensure that standard laptop selection works correctly.
• To verify that approval requests are sent to the correct approver.
• To ensure that approved requests move to the procurement process.
• To verify that notifications are generated at the correct stages.
• To identify and fix errors before deployment.

3. Testing Scope

The testing scope includes the Laptop Request Form, Employee Details, Laptop Model Selection, Request Validation, Approval Workflow, Automated Order Creation, Notifications, Request Status, Rejection and Cancellation handling, and Error Handling.

4. Test Environment

Testing is performed in a controlled test environment. Test data includes Employee Name, Employee ID, Department, Laptop Requirement, Standard Laptop Model, Request Reason, Manager/Approver Details, and Delivery Details. Test users include Employee/User, Manager/Approver, IT Procurement Team, and IT Administrator.

5. Test Cases

Test Case ID	Test Scenario	Expected Result	Status
TC01	Submit a new laptop request	Request should be submitted successfully	Pass
TC02	Leave a mandatory field empty	System should display a validation message	Pass
TC03	Select a standard laptop model	Selected model should be saved correctly	Pass
TC04	Enter valid employee details	Employee information should be accepted	Pass
TC05	Send request for approval	Request should reach the assigned approver	Pass
TC06	Approver approves request	Request status should change to Approved	Pass
TC07	Approver rejects request	Request status should change to Rejected	Pass

Test Case ID	Test Scenario	Expected Result	Status
TC08	Send approved request to procurement	Order processing should begin	Pass
TC09	Check notification	Correct notification should be generated	Pass
TC10	Check request status	Latest request status should be displayed	Pass
TC11	Enter invalid data	System should prevent invalid submission	Pass
TC12	Submit a duplicate request	Duplicate request should be identified/prevented	Pass

6. Functional Testing

Functional testing verifies that each feature of the automated procurement system works according to the requirements. The testing includes creating a laptop request, updating request details, validating mandatory information, selecting a standard laptop, approving or rejecting requests, creating procurement orders, sending notifications, and updating request status.

7. Workflow Testing

The complete workflow is tested from beginning to end: Employee Request → Validation → Manager Approval → IT Procurement → Order Processing → Delivery/Completion. Each stage is checked to ensure that the request moves to the correct next stage.

8. Negative Testing

Negative testing verifies how the system handles incorrect inputs. Examples include missing employee information, invalid employee ID, empty mandatory fields, duplicate requests, invalid laptop selection, and unauthorized approval attempts.

9. User Acceptance Testing (UAT)

User Acceptance Testing confirms that the system meets the practical requirements of employees, managers, and the IT procurement team. Users verify that the request form is easy to use, laptop options are clear, the approval process works correctly, notifications are understandable, and request status can be tracked.

10. Defect Identification and Resolution

Any issue identified during testing is recorded as a defect. Each defect is documented with Defect ID, Description, Test Case ID, Severity, Steps to Reproduce, Expected Result, Actual Result, Resolution, and Retesting Status. After a defect is fixed, the related test case is executed again.

11. Testing Results

The major components of the automated laptop procurement process, including request submission, validation, approval, notification, and order processing, are tested. Any defects identified during testing are corrected and retested before moving the project to the deployment phase.

12. Conclusion

The Testing Phase ensures that the Automated Standard Laptop Ordering System is reliable, accurate, and ready for implementation. It helps reduce manual errors, verifies the approval workflow, and improves the efficiency and standardization of laptop procurement.